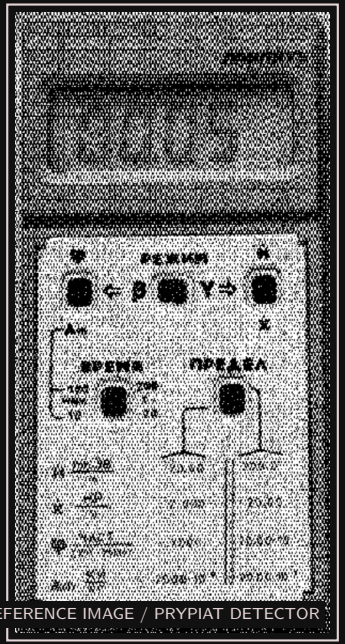
LM-IDNet

Learning normal traffic. Detecting meaningful change.

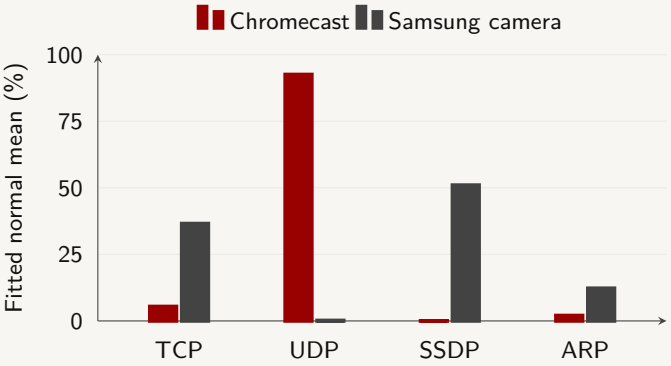
A compact, explainable anomaly detector for IoT network traffic

Fatemeh Mahvari
Supervisor: Professor Mauro Migliardi
Discussion presentation ▪ 19 October 2026



REFERENCE IMAGE / PRYPIAT DETECTOR

A small traffic profile for each IoT device



The question

Can a device's own normal traffic reveal suspicious change?

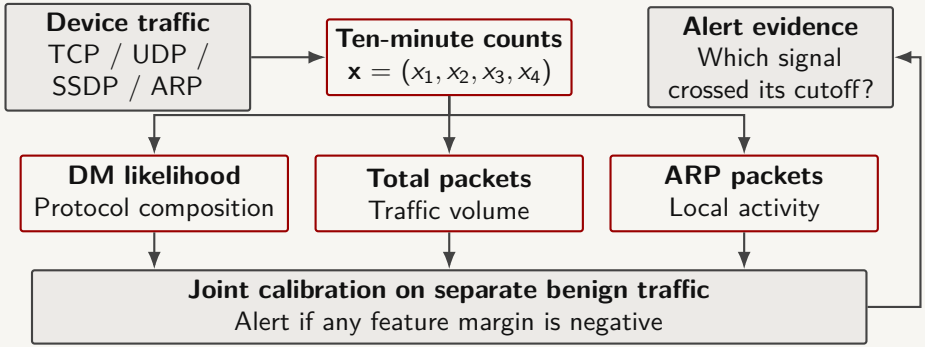
4 exclusive counts
per ten-minute window

Learn from a trusted benign period; flag unusual later behaviour.

A separate normal model captures each device's traffic pattern.

Fitted means from saved normal-only models. UDP excludes SSDP; bars are model expectations.

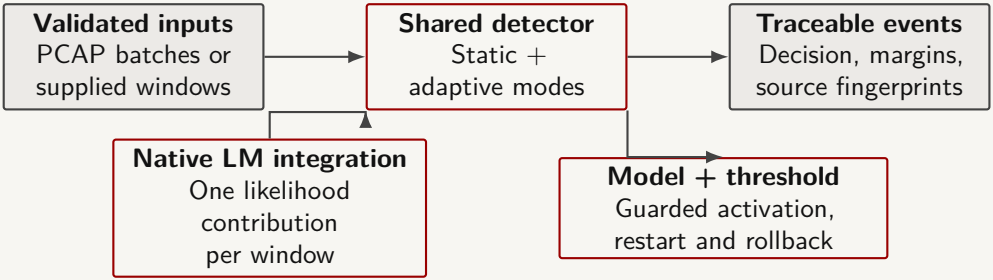
One normal model, three complementary signals



The DM model represents ordinary variation; count tails add complementary evidence.

Normal-only fitting and calibration. The native LM backend evaluates the DM likelihood.

From a numerical method to a usable detector

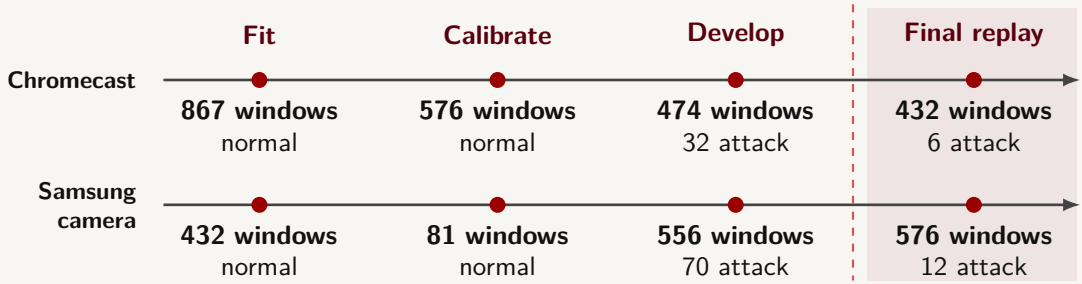


0 / 270 LM versus SciPy decision mismatches in the retained D-Link benchmark. Maximum absolute score difference: 1.09×10^{-5}.

The contribution spans numerical integration, detection, and reliable operation.

Retained backend benchmark and runtime/state contracts; timings are in reference slide R2.

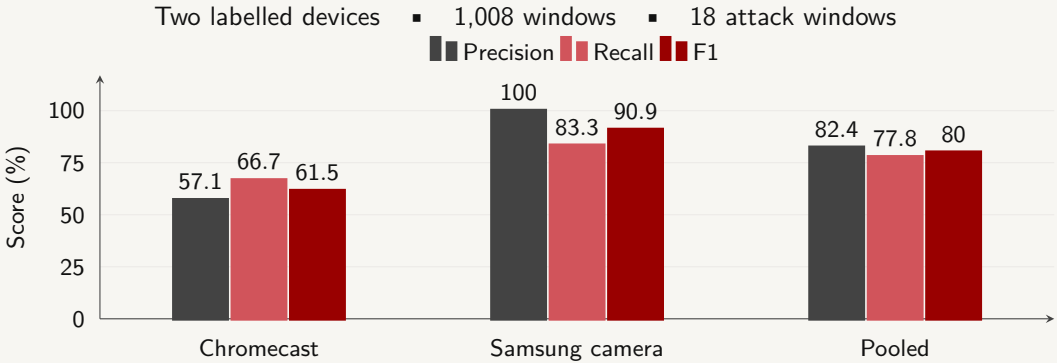
Normal learning first; labelled evaluation later



Final replay: fixed settings, saved normal-only starting pairs, labels only for metrics.

Original reference partitions. Timeline shows role order, not elapsed time. D-Link is a numerical workload without attack labels.

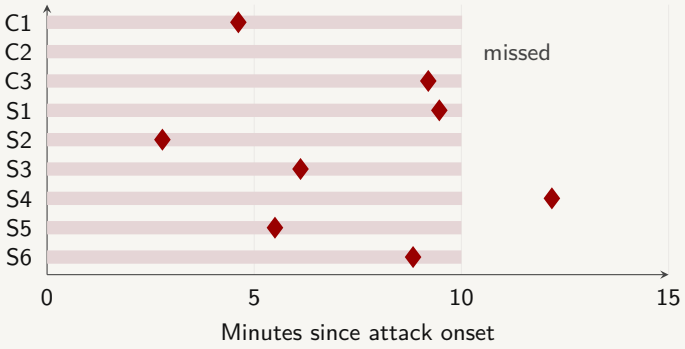
80% pooled F1 on the final partitions



14/18 attack windows flagged; 3/990 benign windows flagged (0.30% FPR).

Final results. Pooled metrics use summed confusion counts. All three selected modes make the same decisions.

Eight of nine attacks detected; seven by attack end



First alert
Annotated attack

Median delay
among detected attacks

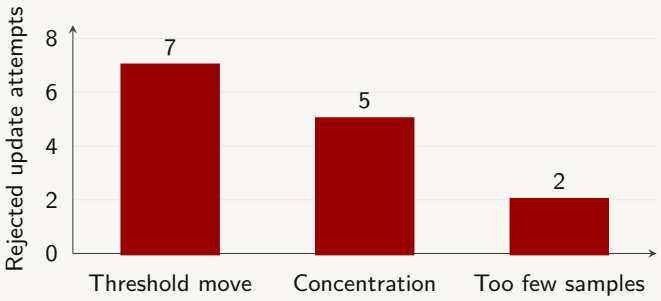
6.9 min
Chromecast

7.5 min
Samsung camera

The ten-minute collection interval is part of the detection delay.

Final official attack intervals, aligned to onset; C = Chromecast, S = Samsung. Alerts occur at window close.

Conservative updates preserved the starting detector



0 / 14

updates activated

Score before learning.

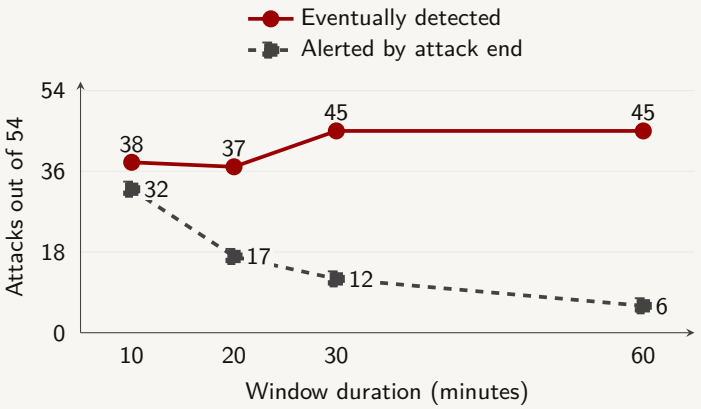
Keep original detection references.

Activate only a complete validated model–threshold pair.

No adaptive gain is established on this final replay.

Seven threshold attempts and seven refit attempts across both devices; paired threshold/refit records counted once.

Window size trades coverage for response time



Delivered default

10 minutes

Static coverage candidate

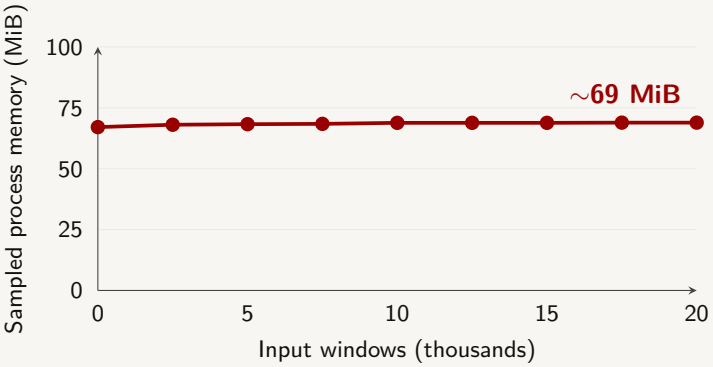
30 minutes

45/54 attacks detected, but only 12/54 alerted by attack end.

Retain ten minutes for earlier response; validate longer windows against the use case.

Separate static development study on matched traffic and 54 attacks; fresh fits with a common normal split across durations.

A measured path towards edge operation



Local Docker simulation
1 CPU / 512 MiB

5.079 s
worst observed processing
including a rejected refit

20,000 supplied windows
16,000 scored; 4,000 missing

Processing has observed headroom; physical gateway performance remains unmeasured.

Retained x86_64 synthetic replay; no update promoted. Processing starts after input arrival; collection delay is separate.

A complete detector with a clear operating tradeoff

4 counts

Compact statistical core

Normal-only DM learning, native likelihood, explainable count signals.

80% F1

Traceable final evidence

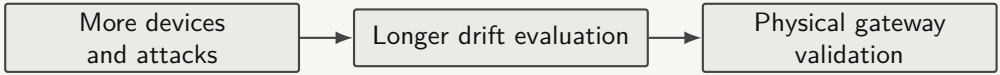
Two labelled devices, 1,008 windows, full decisions and update records.

∼69 MiB

Operational software

Shared batch/streaming detection, guarded state, restart and rollback.

Evidence limits 18 final attack windows; count-preserving attacks can be missed; useful conservative learning remains unproven.



LM-IDNet turns a compact traffic model into a measured, reproducible detection system.

Thank you

Questions and discussion

LM-IDNet ▪ Fatemeh Mahvari

Reference slides follow: model, backend, counts, guards, duration, sources.

R1 / What is fitted, and what triggers an alert?

DM composition model $\alpha_k > 0$, $\alpha_0 = \sum_k \alpha_k$, $N = \sum_k x_k$

$$\ell(\mathbf{x}) = \log \frac{N!}{\prod_k x_k!} + \log \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + N)} + \sum_k \log \frac{\Gamma(\alpha_k + x_k)}{\Gamma(\alpha_k)}.$$

Three signals $\ell(\mathbf{x})$, $-\log(1 + N)$, $-\log(1 + x_{\mathrm{ARP}})$.

Fit-only medians and IQRs scale each signal. The minimum scaled signal is calibrated jointly at the 1% lower quantile on separate benign windows.

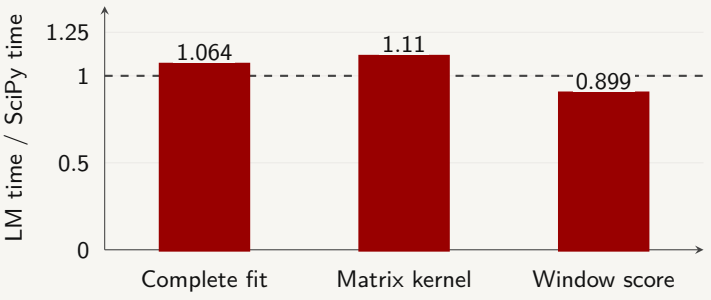
The event score is the minimum signed margin from the resulting cutoffs:

$$s = \min(m_{\mathrm{DM}}, m_N, m_{\mathrm{ARP}}), \qquad \text{alert if } s < 0, \qquad \text{severity} = \max(0, -s).$$

During updates, original DM and count references remain active detection checks.

Constant fit features use unit scale. A count-tail breach can decide early. Severity is not an attack probability.

R2 / Native accuracy and speed are separate claims



D-Link workload

846 × 4 fit matrix

270 scored windows

Numerical agreement

3.86×10^{-7} max. relative α difference

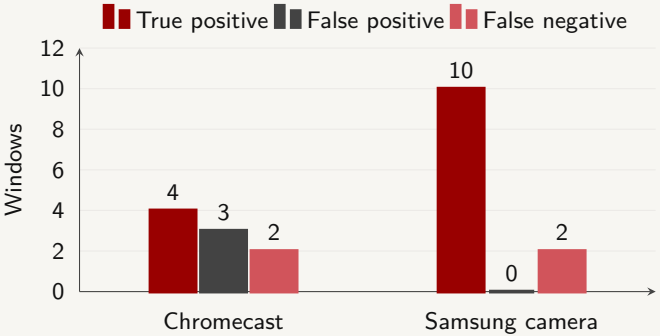
1.09×10^{-5} max. absolute score difference

0/270 decision mismatches

LM is slightly faster for single-window scoring on this workload; fitting is slower.

Retained D-Link benchmark, five timing batches. Ratio 1 = equal time; smaller = faster. No fresh performance run.

R3 / What sits behind the final percentages?



Pooled confusion counts

TP = 14 FP = 3

FN = 4 TN = 987

Precision = 14/17 = 82.35%

Recall = 14/18 = 77.78%

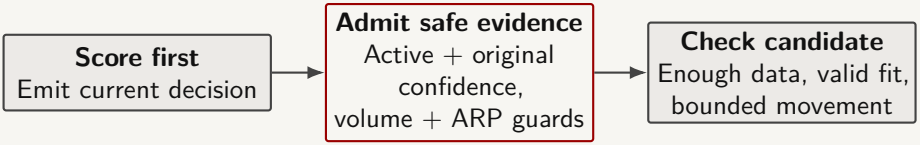
F1 = 28/35 = 80.00%

FPR = 3/990 = 0.30%

Window recall and attack-interval coverage answer different questions.

True negatives: Chromecast 423, Samsung 564. Any attack overlap labels a window positive; all 1,008 windows match labels.

R4 / What does conservative learning permit?



$$1/1.25 \leq \alpha_0'/\alpha_{0,t} \leq 1.25 \qquad \text{concentration ratio}$$

$$\max_k |p_k' - p_{k,t}| \leq 0.05 \qquad \text{protocol proportions}$$

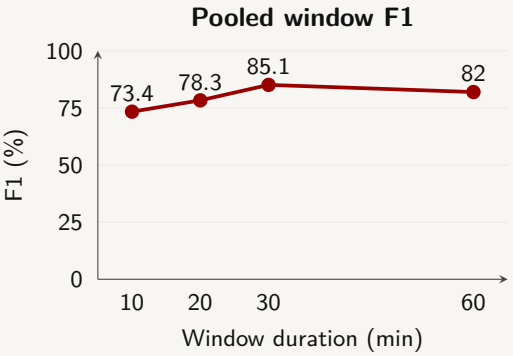
$$|\tau' - \tau_t| \leq 2\,\mathrm{IQR}_0 \qquad \text{threshold movement}$$

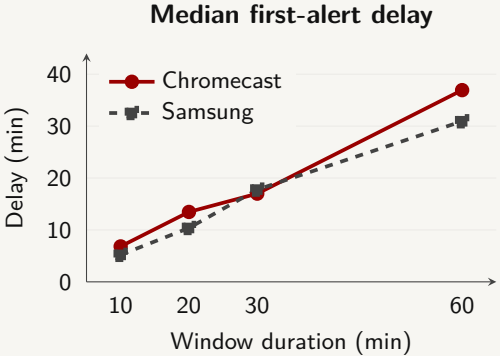
Defaults: 288-window buffer; attempt every 144 scored windows; 144 accepted windows required for periodic fitting.

Rejecting a candidate retains the complete active pair; accepted pairs support rollback.

Per-update guards do not establish immunity to stealthy attacks or unlimited cumulative drift. Final replay promoted no candidates.

R5 / Duration sensitivity beyond the headline





F1 changes with label granularity; median delay is conditional on detection.

Static development study. Samsung uses May 28–29 to fit and May 30–31 to calibrate at every duration. Lines join measured settings.

R6 / Scientific basis and evidence trail

Statistical learning

Minka, *Estimating a Dirichlet distribution*, 2000 / revised 2012.
Fixed-point estimation of the DM parameters.

Native likelihood method

Languasco and Migliardi, *Computational Statistics* 38, 2023.
doi: 10.1007/s00180-022-01311-7
Al-Haj Baddar, Languasco and Migliardi, *IEEE TPAMI* 47(2), 2025.
doi: 10.1109/TPAMI.2024.3489645

Project evidence

Final protocol, results and events: `reports/final_evaluation/`
Duration study: `reports/window_size/`
Backend benchmark: `reports/D-LinkDayCam5/`
Supplied-window simulation: `reports/edge/`

Speaker notes map every numerical slide to its retained source files.